# Correlation Between Income Status and Air Quality in Hamilton: How air contaminants from industrial activities disporportionately impacts low income households *

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This paper reports on an analysis of air quality in Hamilton, Ontario, and its relationships with income disparities and steel plant facilities. Data was obtained from the Hamilton Census Data including Open Hamilton and Statistics Canada.

*Keywords*: air quality, income, steel plants, Hamilton, spatial analysis, environmental justice

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*Paper submitted to complete the requirements of ENVSOCTY 4GA3 Applied Spatial Statistics. Contains public sector data made availabe under the City of Hamilton's Open Data Lisence.

## Introduction

Air pollution is a pressing environmental and public health issue, particularly in urban areas with significantly high industrial activities. Hamilton, Ontario, a Canadian port city situated at the western end of Lake Ontario (Weaver, 2023), is often referred to as the "Steel City" due to the legacy and ongoing presence of steel manufacturing industries in the area (Olalere, 2024). Currently, there are concerns that these steel factories are negatively affecting the health of citizens in surrounding communities due to the air pollutants that these steel plants emit, including sulfur dioxide ($SO_2$), nitrogen dioxide ($NO_2$), and carbon monoxide (CO) (Lui et al., 2014). These concerns raise questions about how socioeconomic factors influence exposure to industrial pollutants. In Hamilton, Ontario, steel production facilities are heavily concentrated in the Bayfront Industrial Area (City of Hamilton, 2023), which is adjacent to downtown neighbourhoods such as Jamesville, Beasly, and Keith, where a considerable portion of the population lives below the poverty line (BTL, 2021; City of Hamilton, 2023). The purpose of this study is to address the research question: Is there a relationship between low-income status and increased exposure to nitrogen dioxide ($NO_2$), sulphur dioxide ($SO_2$), and ozone ($O_3$) emitted from steel mills in Hamilton, Ontario? To answer this question, income statistics and air quality data will be used in a variety of spatial statistical analyses to determine if there is a correlation between household income and the concentration of air pollutants. The results from this study will contribute to a broader understanding of how environmental issues intersect with socioeconomic inequality and aid in the creation of equitable environmental policies in Hamilton.

## Background

### *History of Industries in Hamilton, Ontario*

The city of Hamilton flourished in its early years following the Industrial Revolution, when advanced industries began to emerge. By 1850, approximately one-third of the imports consisting of iron that entered the Burlington canal was routed to the city's foundries for industrial activities (Industrial Hamilton, 2025). The construction of a railway through the shore's inlets in 1852 allowed factories to relocate further outside the heart of the city towards the harbour front and into working class neighbourhoods, creating a centre for industrial work that would persist to this day (Industrial Hamilton, 2025). By 1891, there were several specialized plants in Hamilton that included rolling mills and production plants that made nails, wires, engines, tacks, boilers, bridges, wheels, agricultural machinery and cotton goods. During the 1900s, many companies were formed, including the Steel Company of Canada, the Dominion Foundries and Steel Limited (Dofasco), Stelco Holdings Incorporation, the Chipman Holton Company Limited, the London Machine Company, and many others (Roberts, 1964).

By 1926, the city became the third largest manufacturing center in Canada, serving as a home for over 500 industrial buildings (Industrial Hamilton, 2025). One of the current leading industries includes Dofasco which produces and ships 4.5 million net tons of flat steel across North America yearly (ArcelorMittal, 2025). Currently, Hamilton is known as the "Steel City," with many major industries in the downtown region playing a significant role in Canada's steel production sector (Olalere, 2024).

### *Pollution as a By-product of Steel Production*

The production of steel is a multistep process which contributes a variety of pollutants at several stages. The first step in the steelmaking process is to burn coal in an oven to produce a product termed coke. Next, coke, iron ore, and limestone are melted together in a furnace to produce liquid iron, which is then oxidized in another furnace to create molten steel (ArcelorMittal Dofasco, 2014). During the steel-making process, approximately 500 kg/t of solid wastes such as slags and sludges, are generated. Steel plants also emit air pollutants such as sulfur dioxide ($SO_2$), nitrogen dioxide ($NO_2$), carbon monoxide (CO), and ozone (O3) (Lui et al., 2014). Studies have shown that concentrations of these particles are 50-100% higher in neighbourhoods beside steel plants than in neighbourhoods 5 or more kilometres away from steel plants (Lui et al., 2014).

*Neighbourhood Vulnerability and Income Demographics*

The increasing number of manufacturing industries created many new jobs for labourers in Hamilton, however, with this came a rise in population and a demand for more housing in the downtown area. This pattern persists today, as in 2021 alone, the city of Hamilton increased in population by 6% and increased in total occupied dwellings by 5.3%. Rent and property values both increased in response to the population growth, making it difficult to achieve financial stability in many communities– specifically these downtown neighbourhoods which have seen the most growth and intensification (Hamilton, 2024). From a recent study, it was found that the downtown Hamilton areas are home to the "poorest of the poor" where gross household incomes are 40 to 60 percent lower than the average low-income families (BTL, 2021). 40% of Ward 2's population lives below the poverty line. Eight neighbourhoods located in three downtown wards report more than half of children living in poverty (BTL, 2021).

*Pollution Impacts on Health*

The effects of steel plant pollution on human health and diverse environments have become a more well-studied topic in recent years. In 2023, Jion et al. (2023) discussed that the United States Environmental Protection Agency (US EPA) and the World Health Organization (WHO) deemed that $NO_2$ is a severe pollutant in the atmosphere (Jion et al., 2023). Additionally, $SO_2$ specifically contributes to health issues such as asthma, respiratory irritation, bronchitis, and cancer in humans. Chronic exposure to low levels of CO may impair neurodevelopment and cause a lack of memory, attention, and concentration (Téllez et al., 2006). CO may also disrupt cardiovascular processes, leading to high blood pressure, arrhythmias, and electrocardiographic signs of ischemia (Levy, 2015). Extended exposure to high levels of O3 may have short-term effects on the respiratory and cardiovascular systems of humans, as well as long-term effects such as respiratory mortality, asthma in children, and increased respiratory effects on patients with asthma (Nuvolone et al., 2018). Additionally, higher O3 concentrations may induce transient alterations in lung function, respiratory symptoms, and airway inflammation, which may worsen in the presence of other environmental variables (Lippmann, 1989).

Steel plant pollution also has a considerable effect on the environment. $NO_2$ and $SO_2$ are poisonous and pose considerable risks to biodiversity, and a mixture of these two pollutants is especially harmful to plants (Jion et al., 2023). $SO_2$ contributes to acid rain, soil acidification, and ozone depletion, while $NO_2$ is harmful to vegetation as it decreases crop production and plant growth efficiency. Both $NO_2$ and $SO_2$ significantly accelerate global warming and eutrophication processes on Earth and in water systems. Elevated atmospheric CO levels may disrupt the hormonal balance and stress response mechanisms of plants (Wang & Liao, 2016). Higher CO concentrations may also cause leaf curling, yellowing, and inhibited growth (Zimmerman et al., 1933). Elevated concentrations of O3 are a prominent stressor within natural ecosystems, as it has been well documented that these increases impact soils, biota, and ecological processes (Grulke & Heath, 2020). O3 may decrease crop yields as it affects the carbon, nutrient, and hydrological dynamics of individual plants, especially when coupled with other stresses such as prolonged drought and excess nitrogen deposition (Grulke & Heath, 2020; Krupa et al., 1995).

Additionally, there is evidence of a relationship between negative health effects and an individual's housing proximity to a steel plant. In 1989, a study by Pope concluded that there was an increase in hospital admissions, specifically for respiratory illnesses, during the years that a steel plant was operating in Utah Valley (Pope, 1989). This research demonstrated that the emissions from the steel plant likely induced respiratory illnesses in the residents of the area (Pope, 1989). Further research has been conducted in later years, such as the 2014 paper by Liu, et al, which investigated the effects of steel mills on human cardiovascular physiology in Sault Ste. Marie, Ontario, Canada (Lui et al, 2014). The results of this study suggest that there is a correlation between higher heart rates for individuals who live near a steel plant. This is concerning as a frequently high heart rate could lead to coronary artery disease or cardiovascular mortality (Lui et al, 2014).

*Relevant Research in the Field*

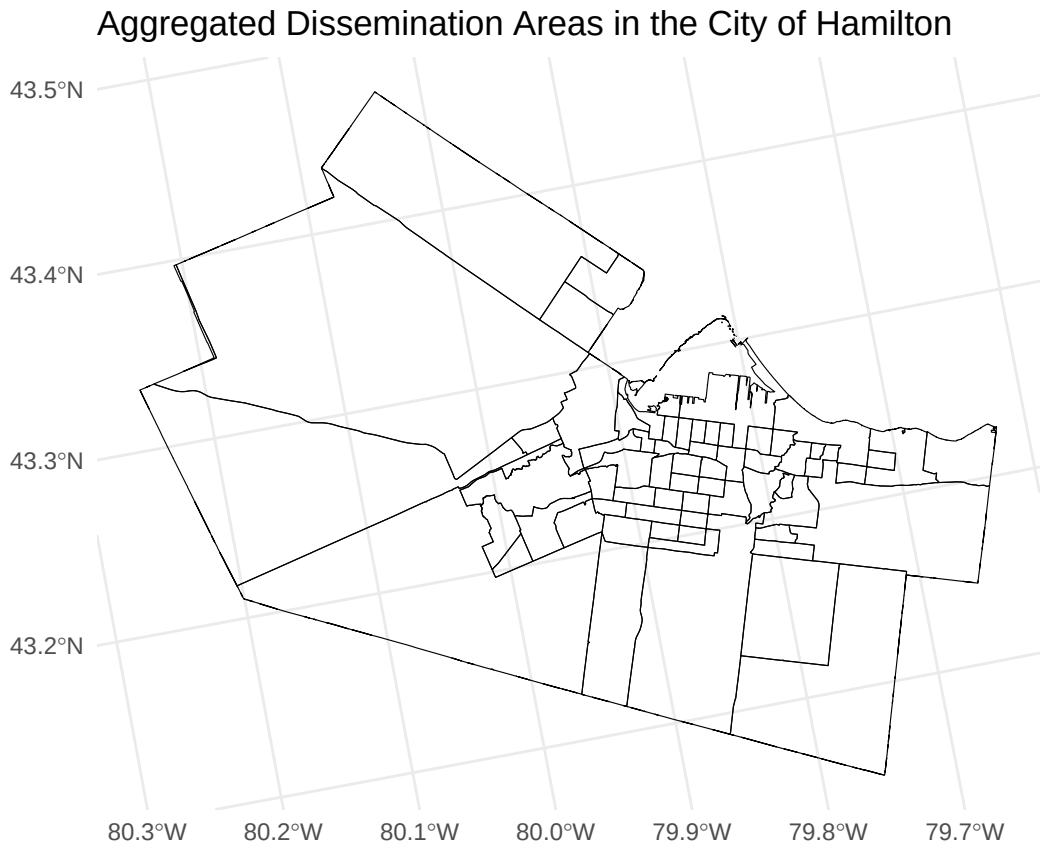
Currently, there is a notable gap in the research investigating the relationship between household income and air pollution in Hamilton, Ontario. In the literature, studies have researched more broadly the effects of urbanization on the natural habitat and the history of the steel industries in Hamilton (Shaw, 2012). More generally, the effects of $NO_2$, $SO_2$, and O3 pollution on the environment and humans have been more well-researched. However, little to no studies have analyzed air pollution data in Hamilton. This research would be very useful for looking at air pollutant concentration changes over time, air pollution effects on humans, and what solutions can be devised to solve these pollution issues.

## Objectives

This study focuses on examining income demographics as a proxy for understanding the health outcomes linked to steel plant pollution in Hamilton. Our research question is, is there a correlation between low-income status and exposure to pollution from the steel mills in Hamilton, Ontario? This research provides insight into the potential sources of health conditions affecting residents living near the industrial area of Hamilton, Ontario. Studying socio-economic relations to health and correlated environmental factors can help ensure that potential harm is reduced in the future. For example, if there is a positive correlation between socioeconomic status and air pollution, targeted mitigation effects can be implemented that suit the communities this study serves. If there is evidence of harm, pollution policies should be strongly enforced, resulting in better human health and fewer environmental contaminants.

## Study Area

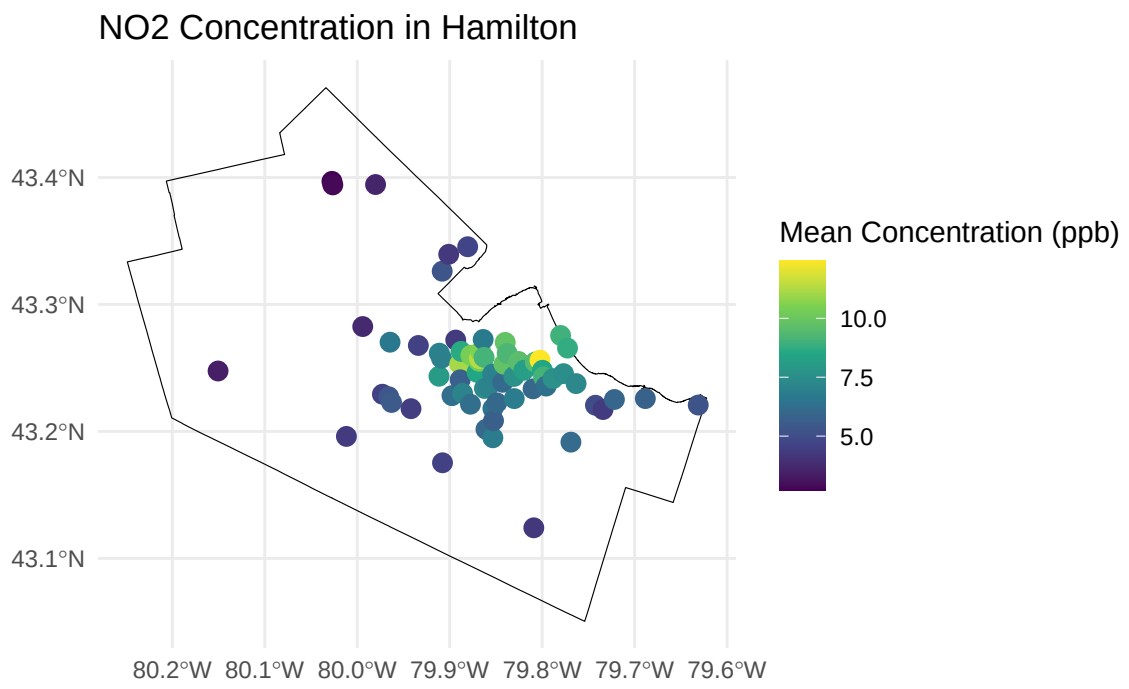
This study is conducted in the city of Hamilton, Ontario, Canada, which has a latitude of $43^\circ\,15'\,19.085''$ North, and a longitude of $79^\circ\,52'\,23.165''$ West. While the data used in this study is related to air monitoring sites, which are located around Hamilton, the source of the pollutants are within the industrial area of the city. The City of Hamilton's Bayfront Industrial Area is a 1,607-hectare area, which occupies several manufacturers, primarily steel plants. These steel plants are on the shore of the Hamilton Harbour (City of Hamilton, 2023). The city is much larger than this 1,607-hectare area, as seen by the "Aggregated Dissemination Areas in the City of Hamilton" map, and occupies several residential neighborhoods to the south of the industrial area.

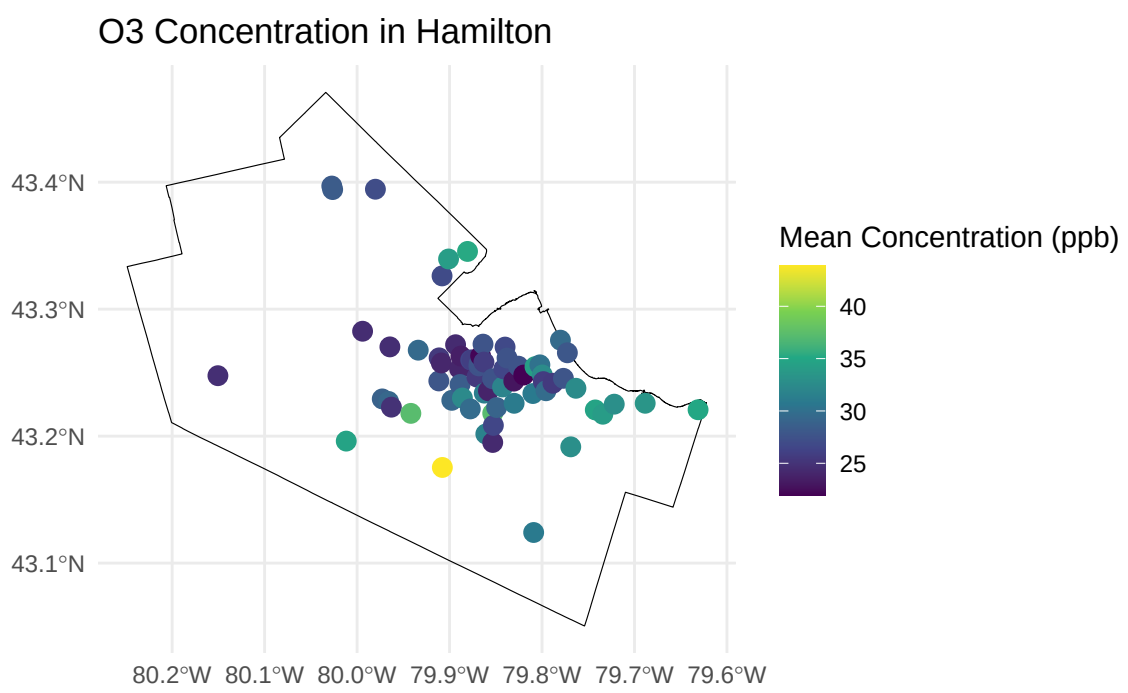


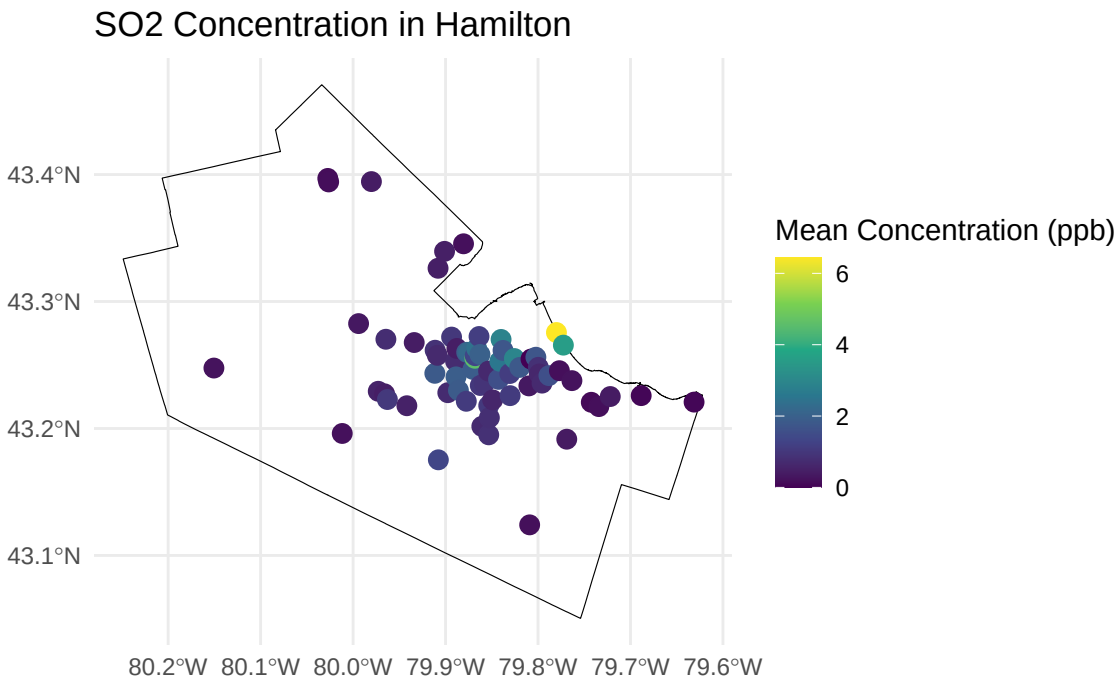

**Data**

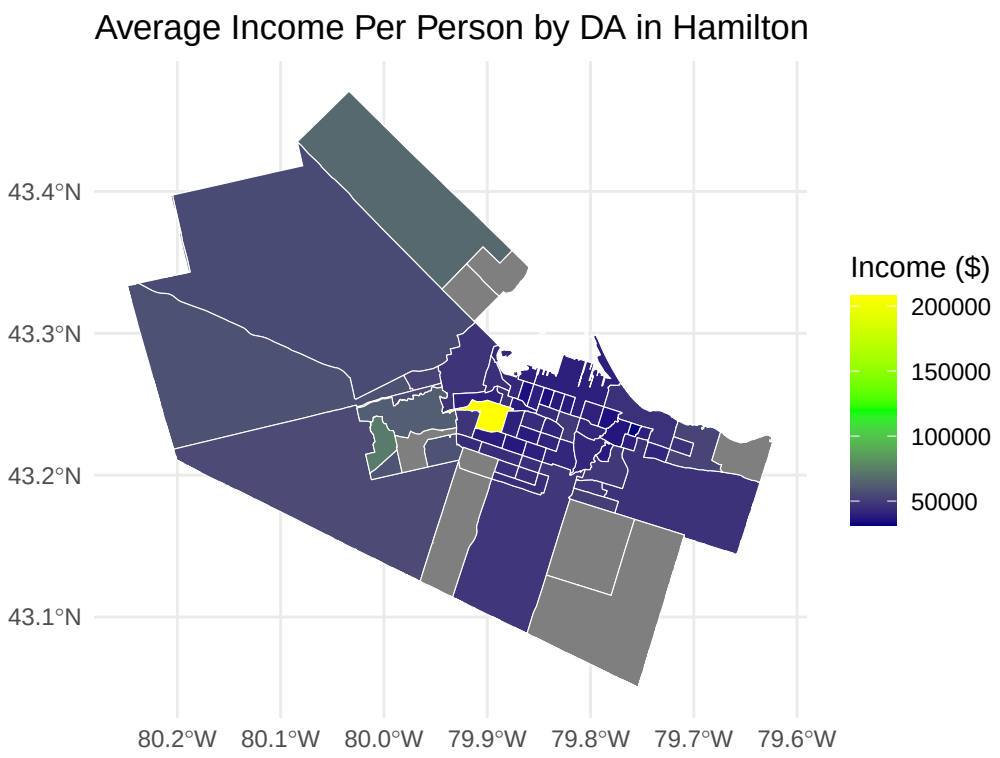
For this project, income statistics are sourced from the 2021 Canadian Census of Population, a series of cross-tabulations based on census information from the Government of Canada. It is displayed at the level of the aggregated dissemination areas. The air quality data and aggregated dissemination area boundaries are sourced from Open Hamilton, the City of Hamilton's publicly available data hub. The air quality data is sourced from a University of Toronto project between February 2022 and May 2023, which measured ground level ozone ($O_3$), nitrogen dioxide ($NO_2$), and sulphur dioxide ($SO_2$).

*Raw data mapped*





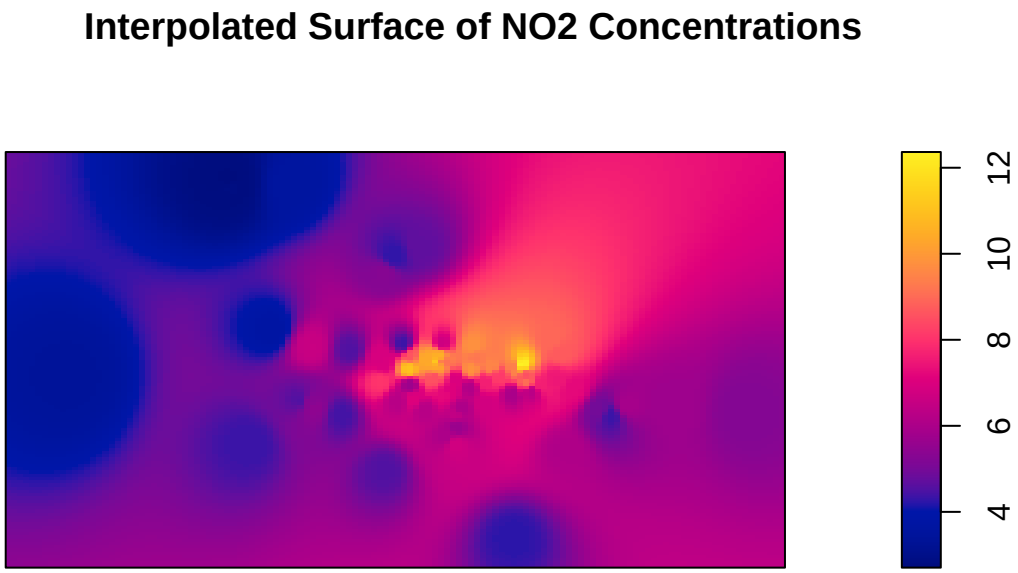




*Formatting our data*

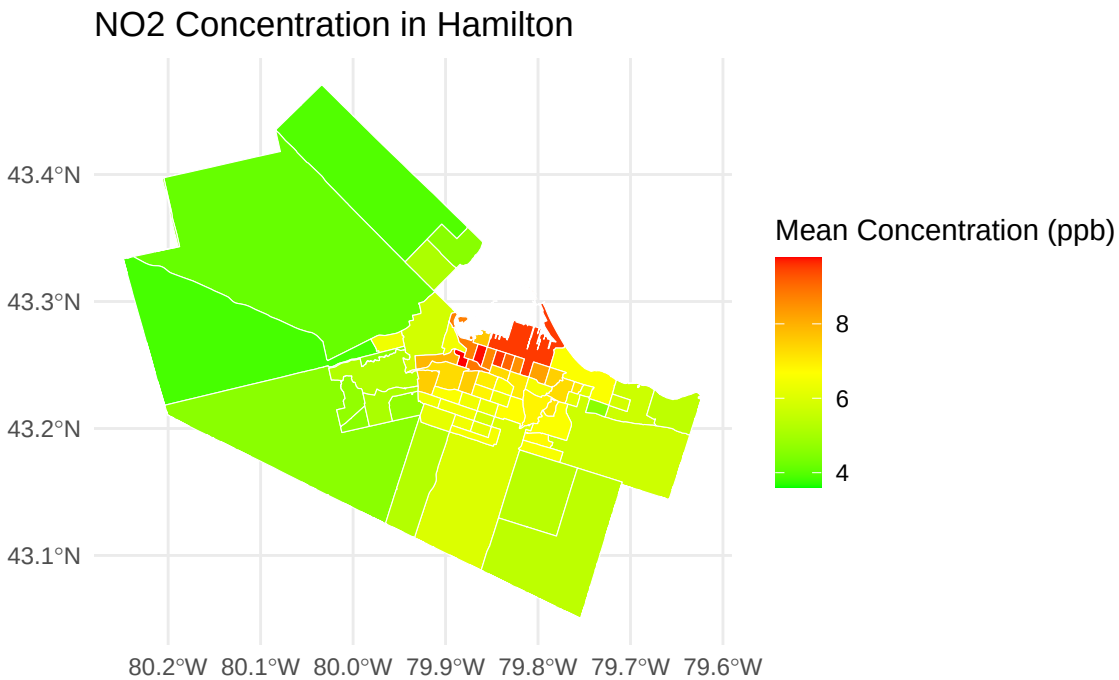
In order to use this air quality data in correlation with the income statistics, which come in values associated with aggregated dissemination areas, we needed to derive one mean concentration of each chemical for every aggregated dissemination area. In order to do this, an interpolated surface of each chemical was created

using inverse distance weighting. An example of this interpolated surface can be seen below.

```
## Warning: attribute variables are assumed to be spatially constant throughout
## all geometries
```




From there, we rasterized each interpolated surface and extracted a mean value for each aggregated dissemination area.




```
## Warning: attribute variables are assumed to be spatially constant throughout
## all geometries
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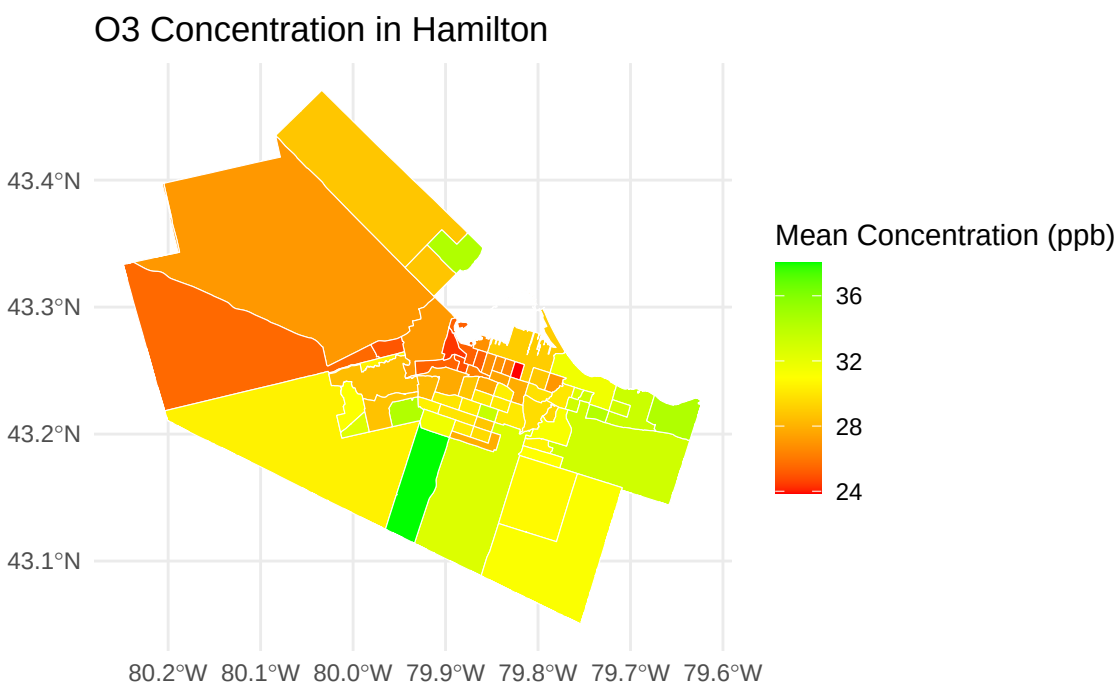
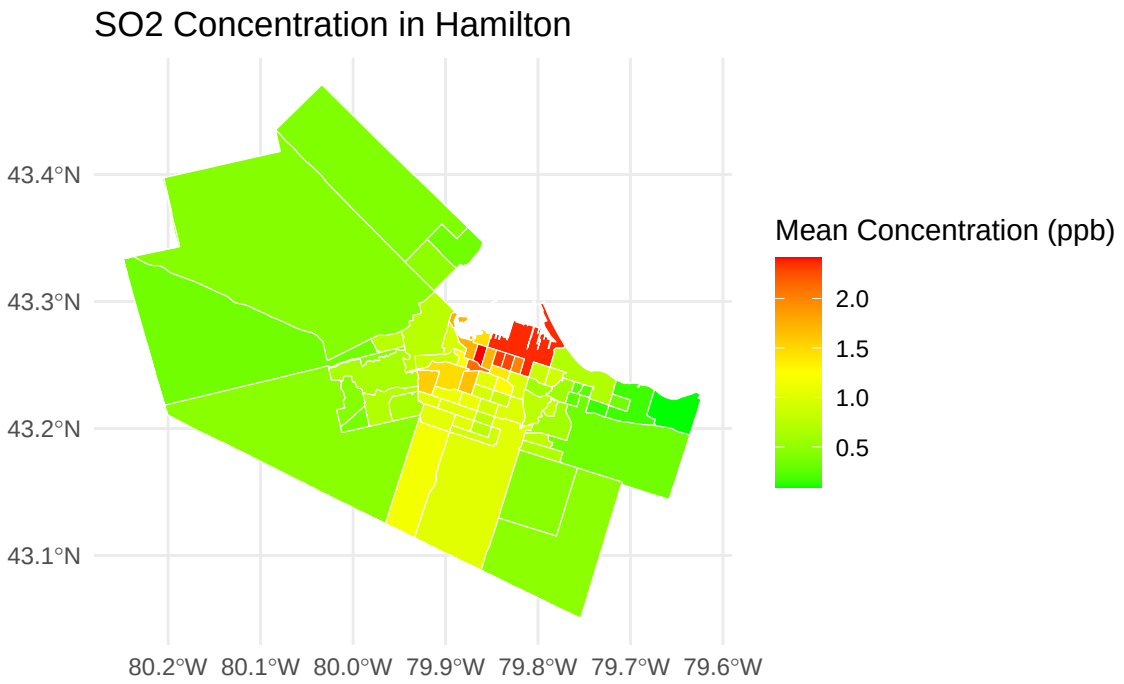
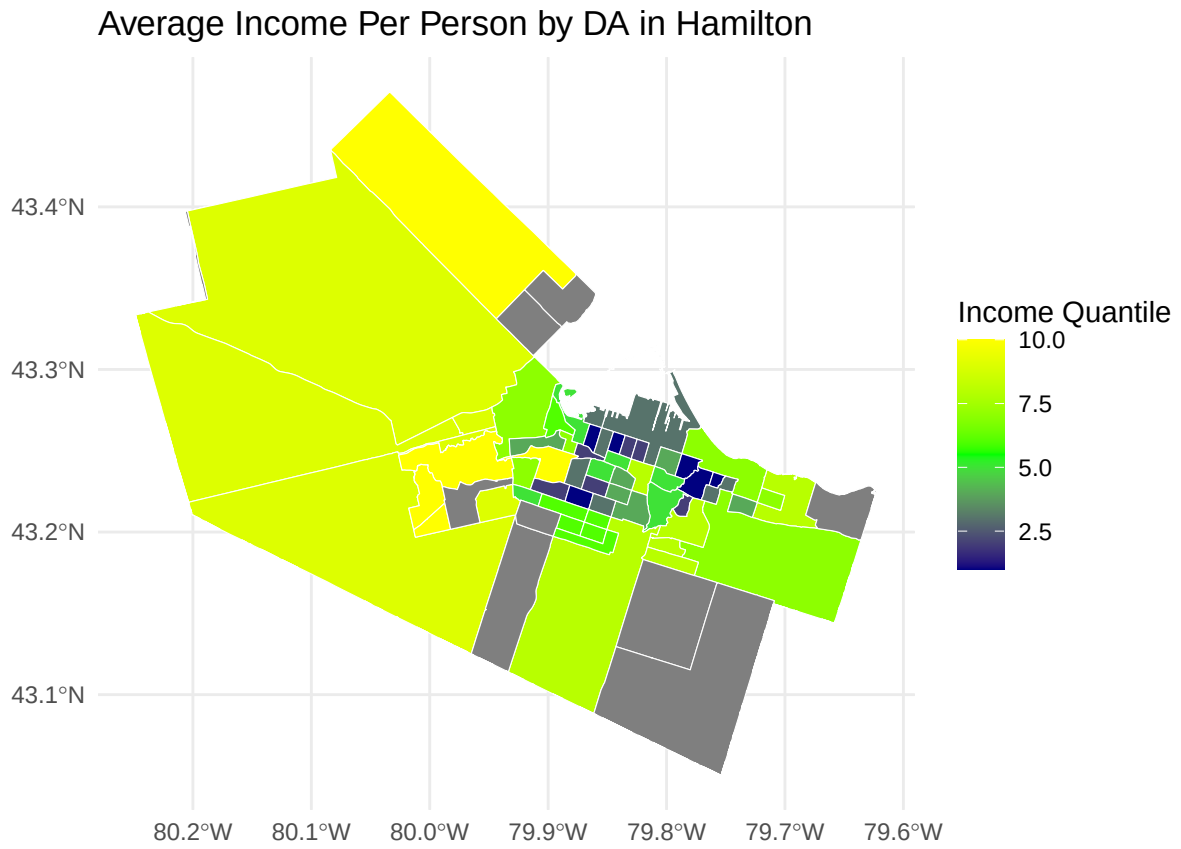



```
## Warning: attribute variables are assumed to be spatially constant throughout
## all geometries
```




We also needed to format the income data to better represent the differences in income status across the city. We did this by using a quantile ranking scheme that divided income status groups into ten. Additionally, part of formatting this data was to remove outliers, including those aggregated dissemination areas for which there were no values or the value was distorting the analysis, including one DA that had an extremely high average income.



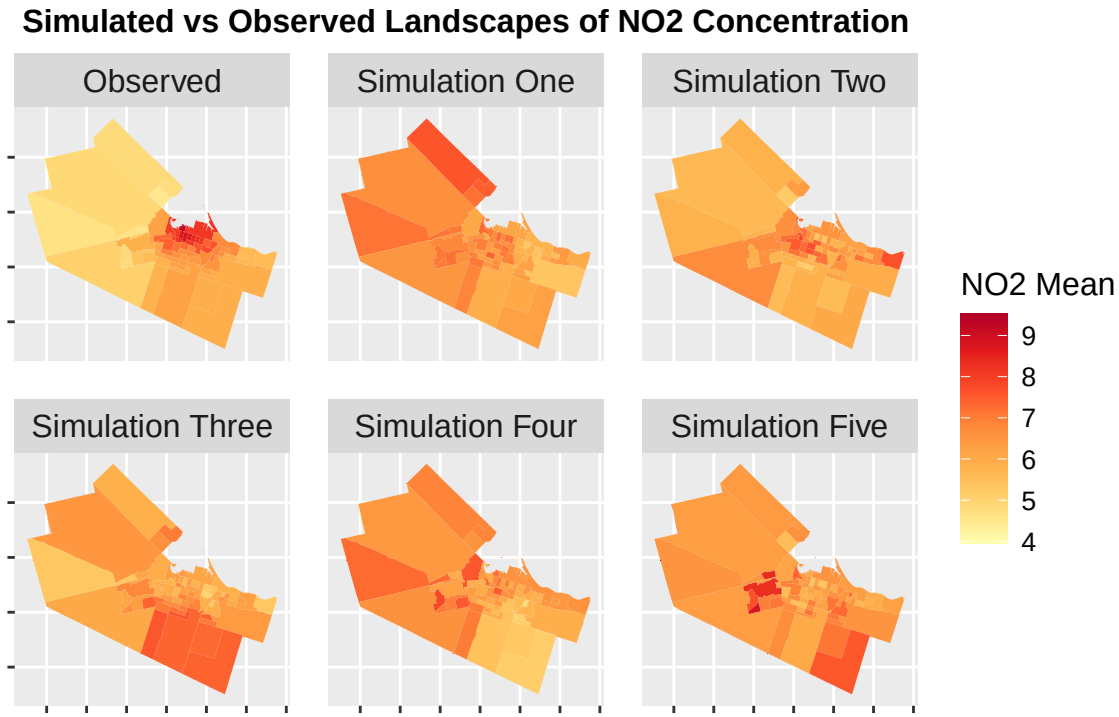

### Methods

In this study, RStudio is used to format and analyse the patterns that occur both in income status and air quality across the City of Hamilton, Ontario, and find a correlation between these variables based on background research and current available data. The independent variables in this case are the level of air contamination from $NO_2$ and $SO_2$, and the concentration of $O_3$. The dependent variable is income status across the City, which is broken down into ten divisions to represent the disparity between aggregated dissemination areas. Inverse distance weighting is used to interpolate a surface of contaminant concentrations, spatial moving averages, simulated landscapes, and Moran's plots are used to find spatial patterns, and the Pearson correlation and regression analyses are used to make plots that find a correlation between contaminants and income status.

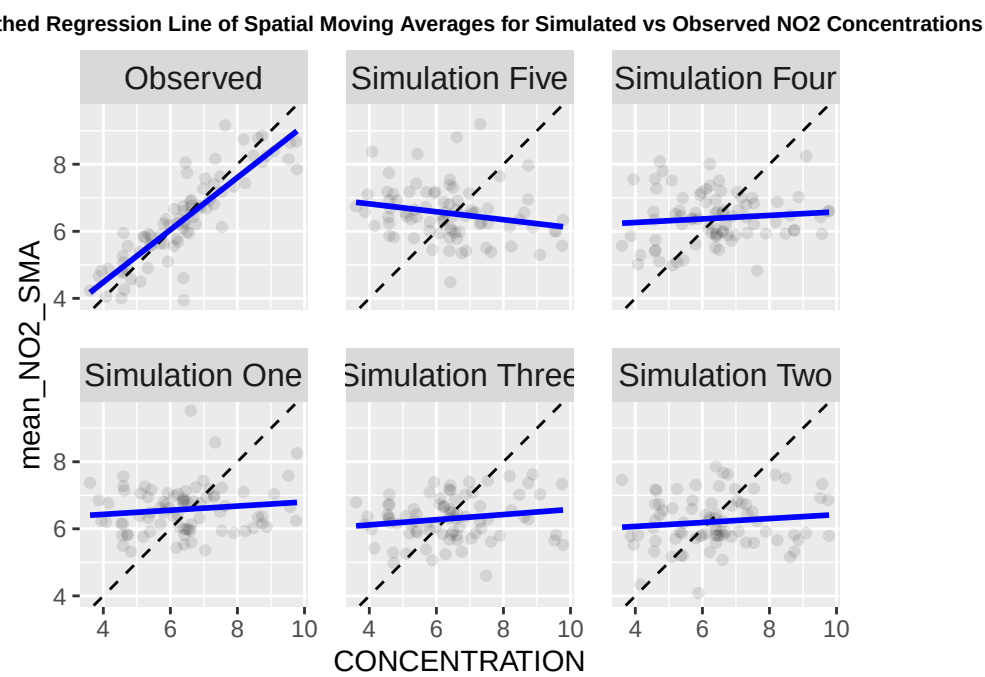
### Analysis

*Is there a pattern in the data?*

We wanted to find out whether the concentration of each chemical was random or stochastic, and therefore, we compared five simulated spatial moving averages to the observed data.




```
## `geom_smooth()` using formula = 'y ~ x'
```




Using simulated landscapes such as these, we are able to find moran's statistics for each pollutant.

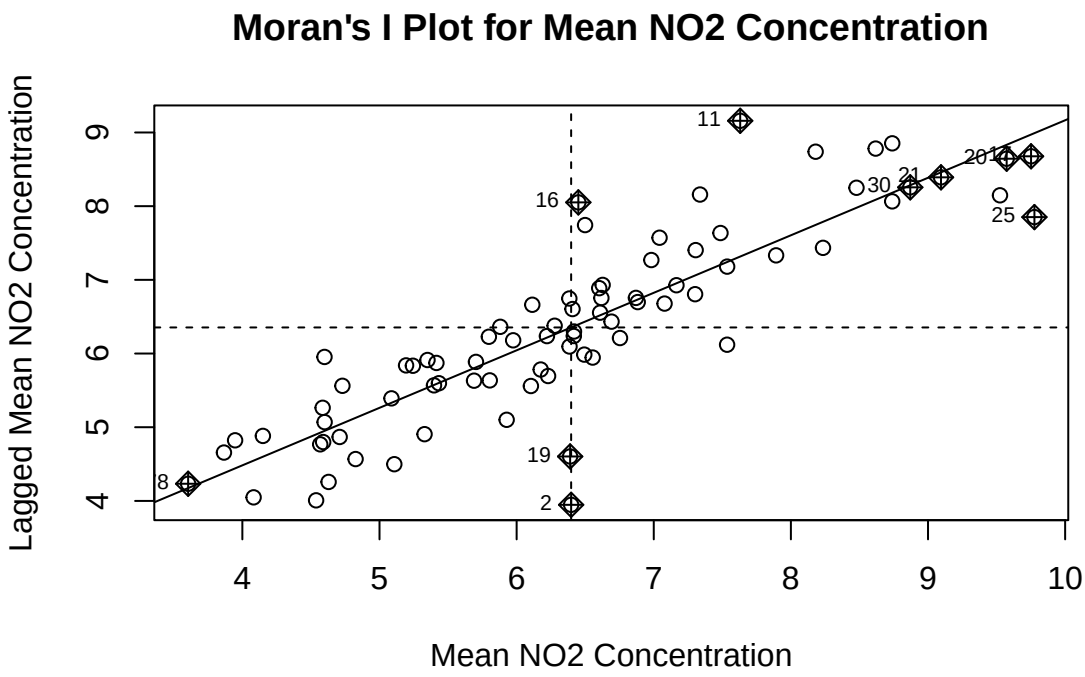
```
##
```

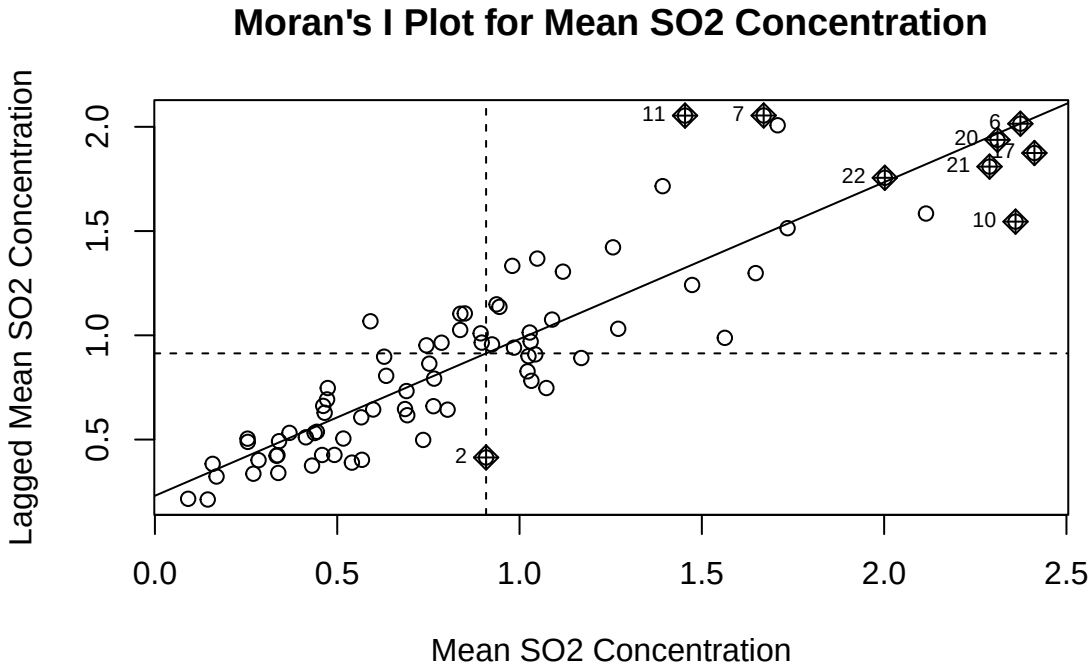
```
##  Moran I test under randomisation
##
## data:  HamDAs_WGS$mean_NO2
## weights: HamDAs_WGS.w
##
## Moran I statistic standard deviate = 11.216, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##       0.780474810      -0.012658228       0.005000659
```

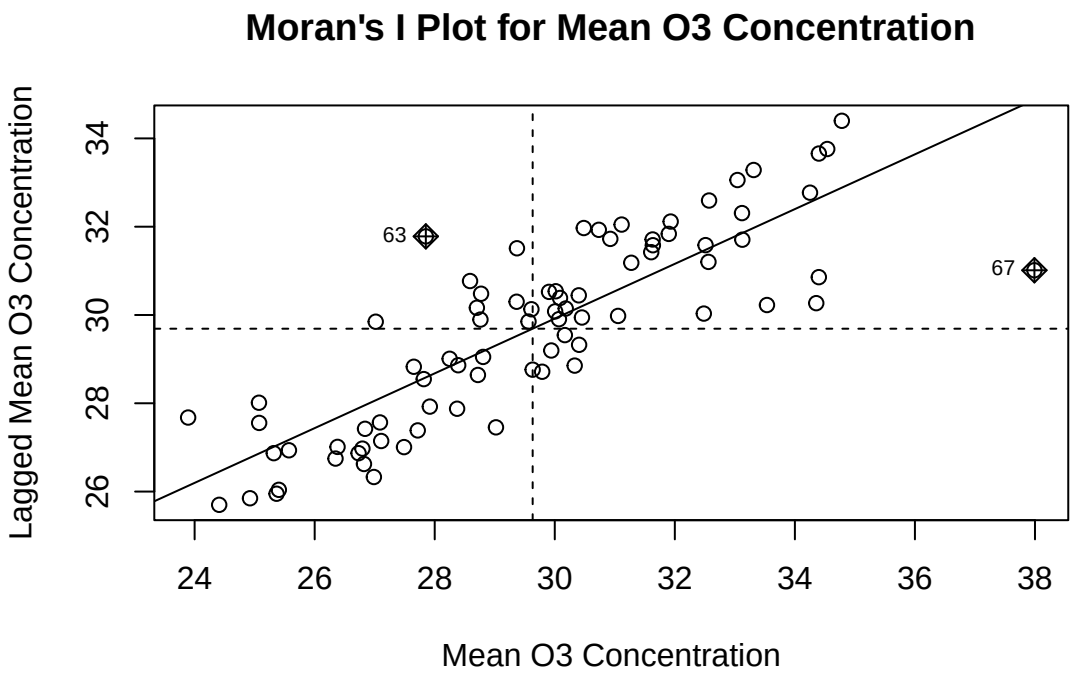
```
##
##  Moran I test under randomisation
##
## data:  HamDAs_WGS$mean_SO2
## weights: HamDAs_WGS.w
##
## Moran I statistic standard deviate = 10.873, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##        0.75199067       -0.01265823        0.00494559
```

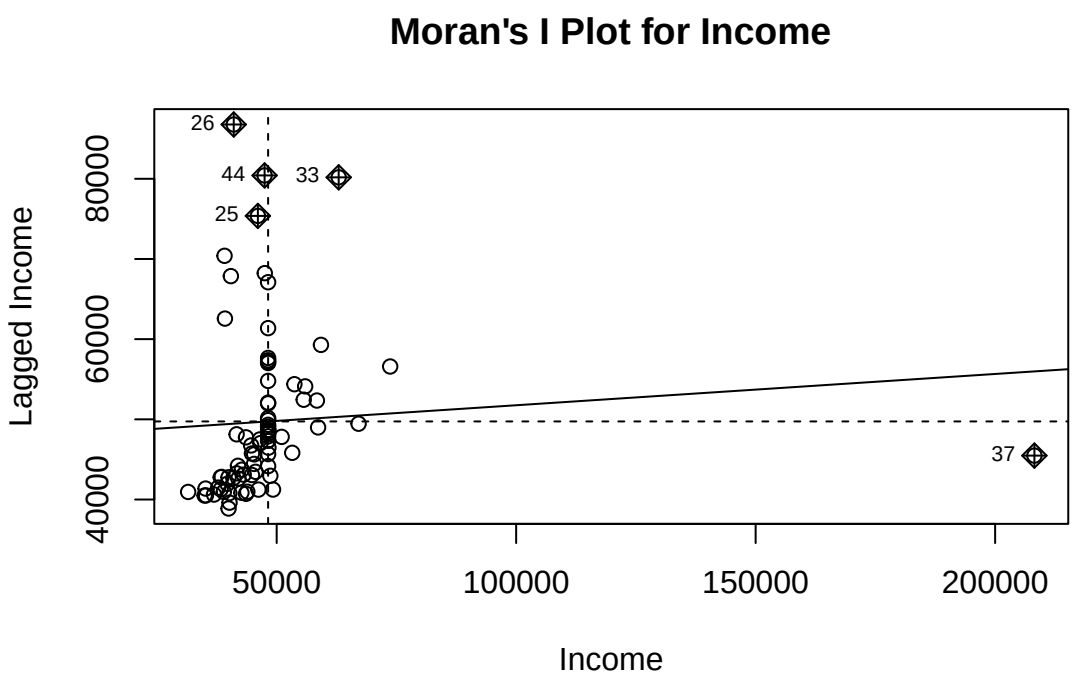
```
##
##  Moran I test under randomisation
##
## data:  HamDAs_WGS$mean_O3
## weights: HamDAs_WGS.w
##
## Moran I statistic standard deviate = 8.9525, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##       0.620232410      -0.012658228       0.004997678
```

```
##
##  Moran I test under randomisation
##
## data:  HamDAs_WGS$C1_COUNT_T
## weights: HamDAs_WGS.w
##
## Moran I statistic standard deviate = 1.4293, p-value = 0.07646
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##       0.039013652      -0.012658228       0.001306956
```

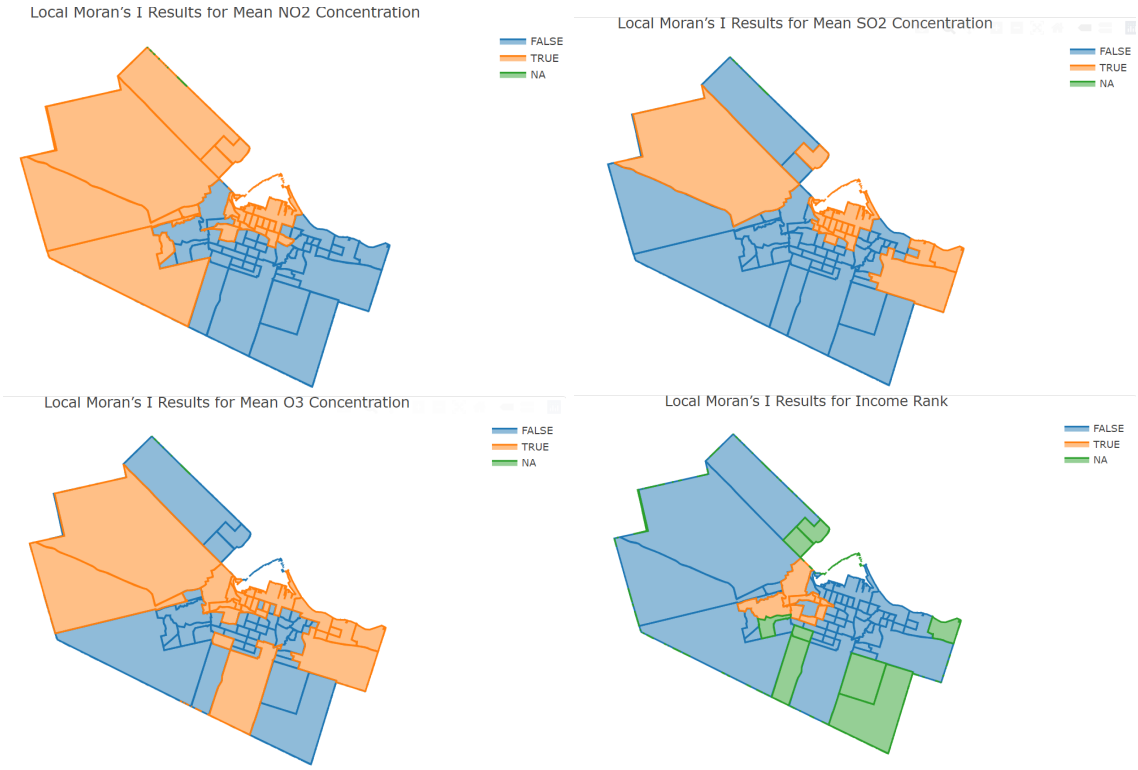

Moran's I Plot for Mean NO2 Concentration
Lagged Mean NO2 Concentration
Mean NO2 Concentration
4
5
6
7
8
9
10
11
16
19
2
8
20
21
30
25


Moran's I Plot for Mean SO2 Concentration
Lagged Mean SO2 Concentration
Mean SO2 Concentration
0.0
0.5
1.0
1.5
2.0
2.5
11
7
6
20
17
21
22
10
2




We can take these tests one step further by mapping the results to visualize the residuals across space.




*Is there a correlation between the data?*

After the data has been formatted, and we have seen how the data relates to itself (random vs stochastic pattern and Moran's tests), we can now move on to understanding how each dataset relates to each other.

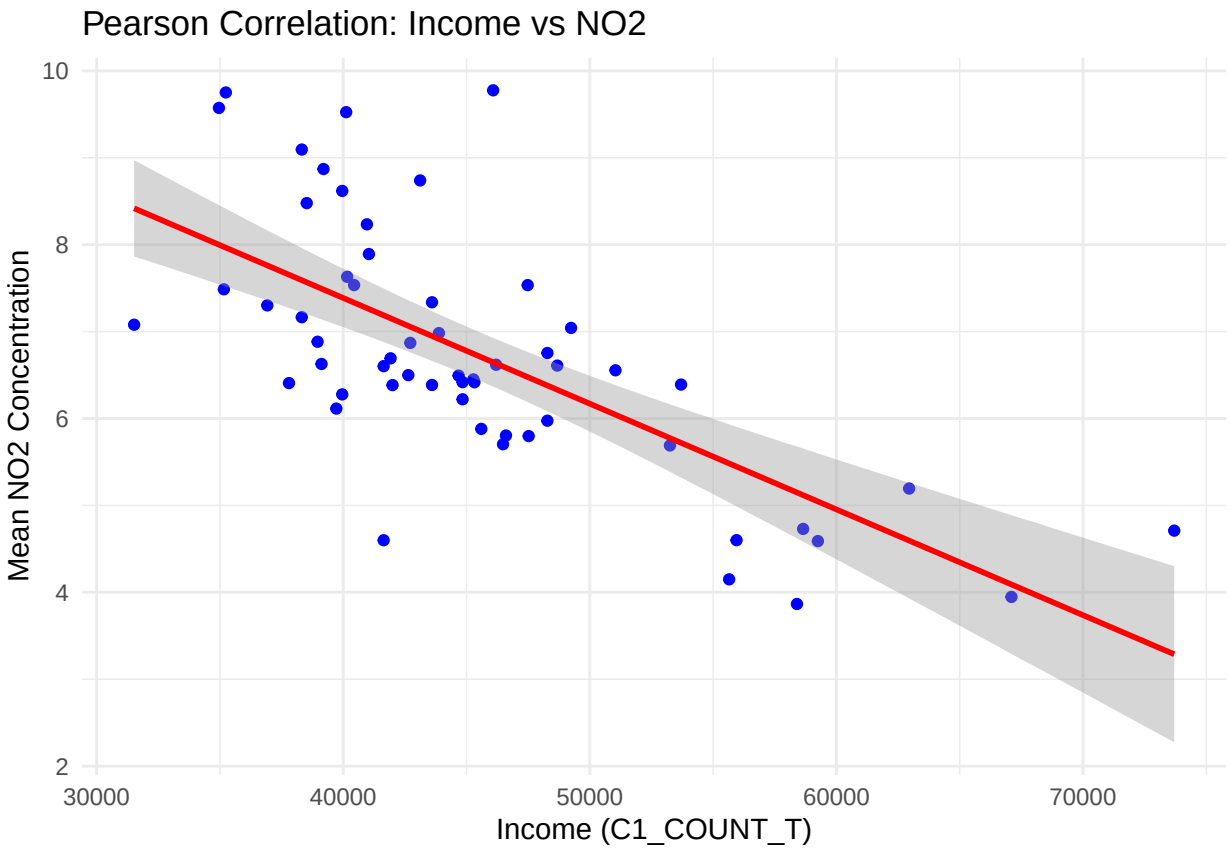
We began by running Pearson tests of correlation for each pollutant compared to the income status data.

```
##
##  Pearson's product-moment correlation
##
## data:  HamDAs_WGScopy$C1_COUNT_T and HamDAs_WGScopy$mean_NO2
## t = -7.0812, df = 55, p-value = 2.777e-09
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.8060976 -0.5243712
## sample estimates:
##        cor
## -0.6905818
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 23 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

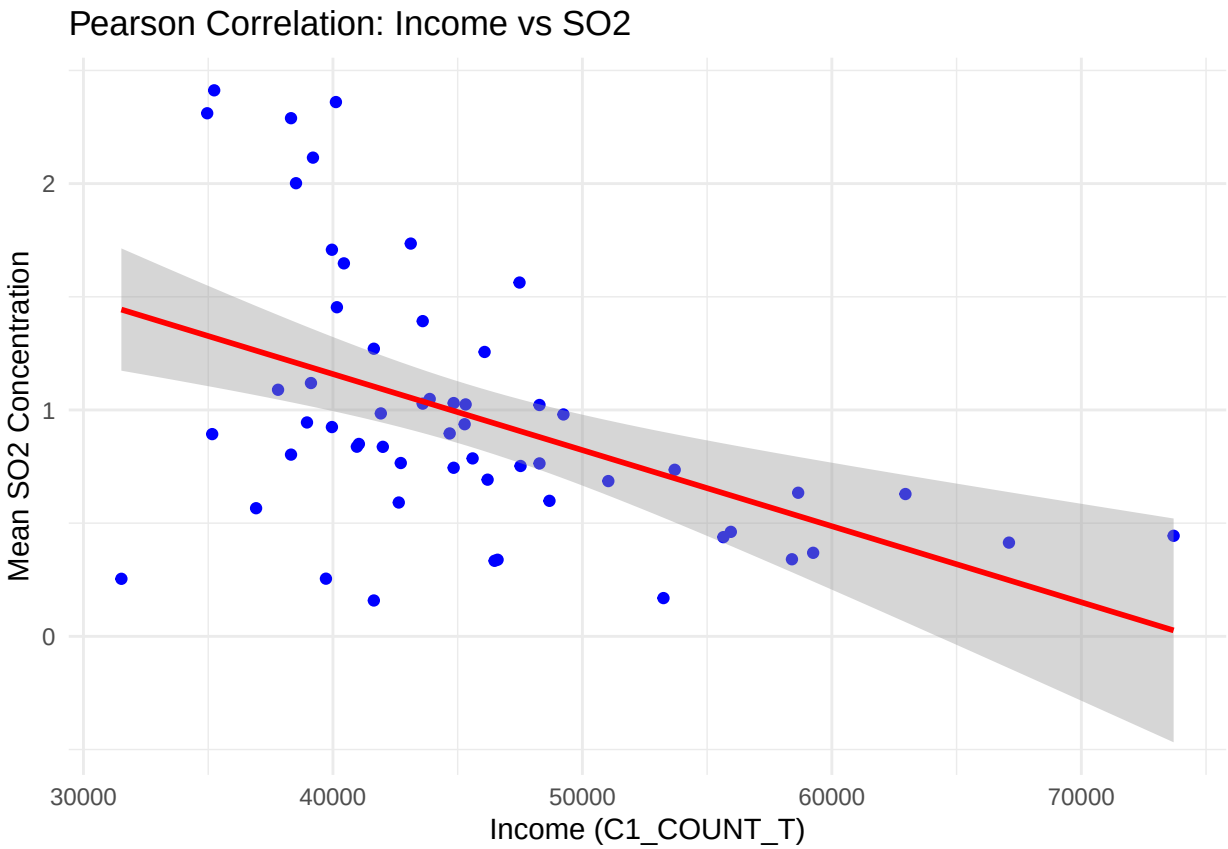
```
## Warning: Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
##
##  Pearson's product-moment correlation
##
## data:  HamDAs_WGScopy$C1_COUNT_T and HamDAs_WGScopy$mean_SO2
## t = -4.0092, df = 55, p-value = 0.0001851
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.6549683 -0.2454034
## sample estimates:
##        cor
## -0.4755616

## `geom_smooth()` using formula = 'y ~ x'

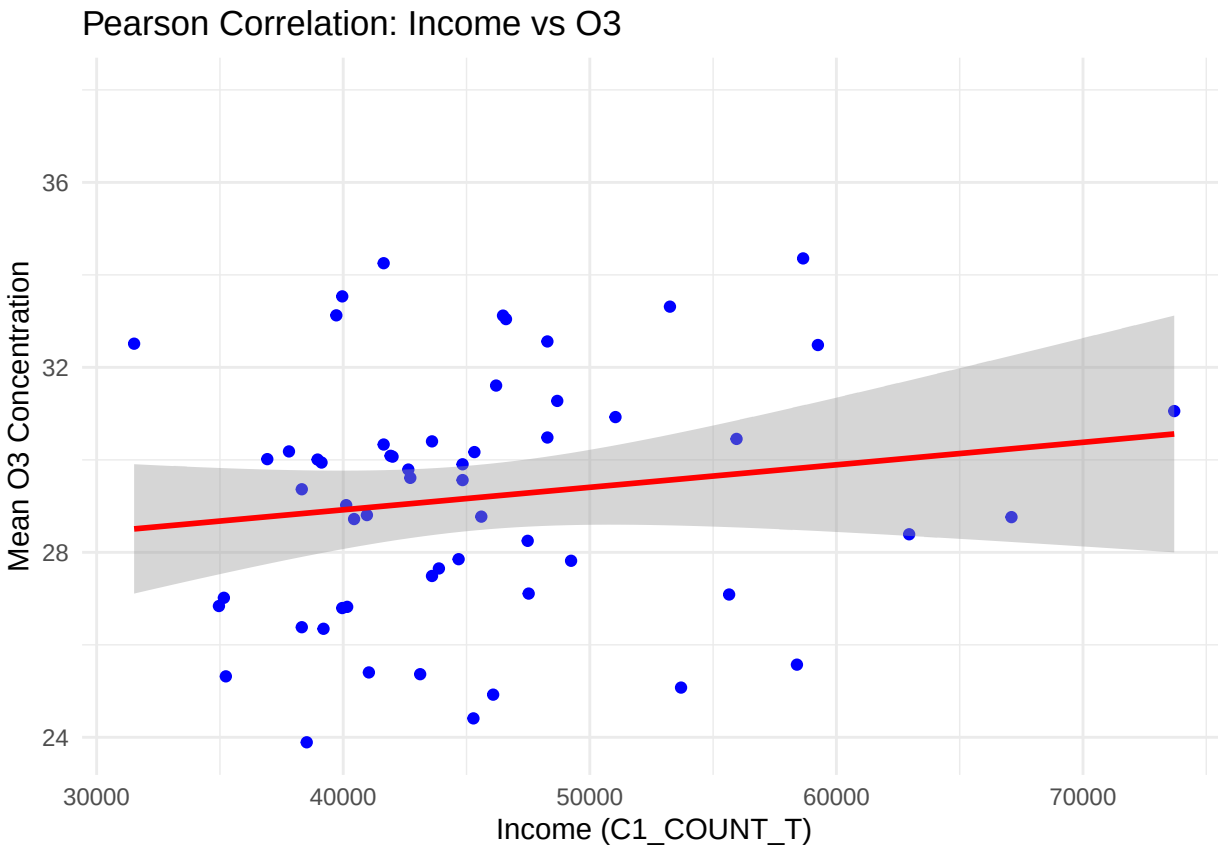
## Warning: Removed 23 rows containing non-finite outside the scale range
## (`stat_smooth()`).
## Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
##
##  Pearson's product-moment correlation
##
## data:  HamDAs_WGScopy$C1_COUNT_T and HamDAs_WGScopy$mean_O3
## t = 1.1211, df = 55, p-value = 0.2671
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.1155934  0.3946721
## sample estimates:
##       cor
## 0.1494764

## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 23 rows containing non-finite outside the scale range
## (`stat_smooth()`).
## Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

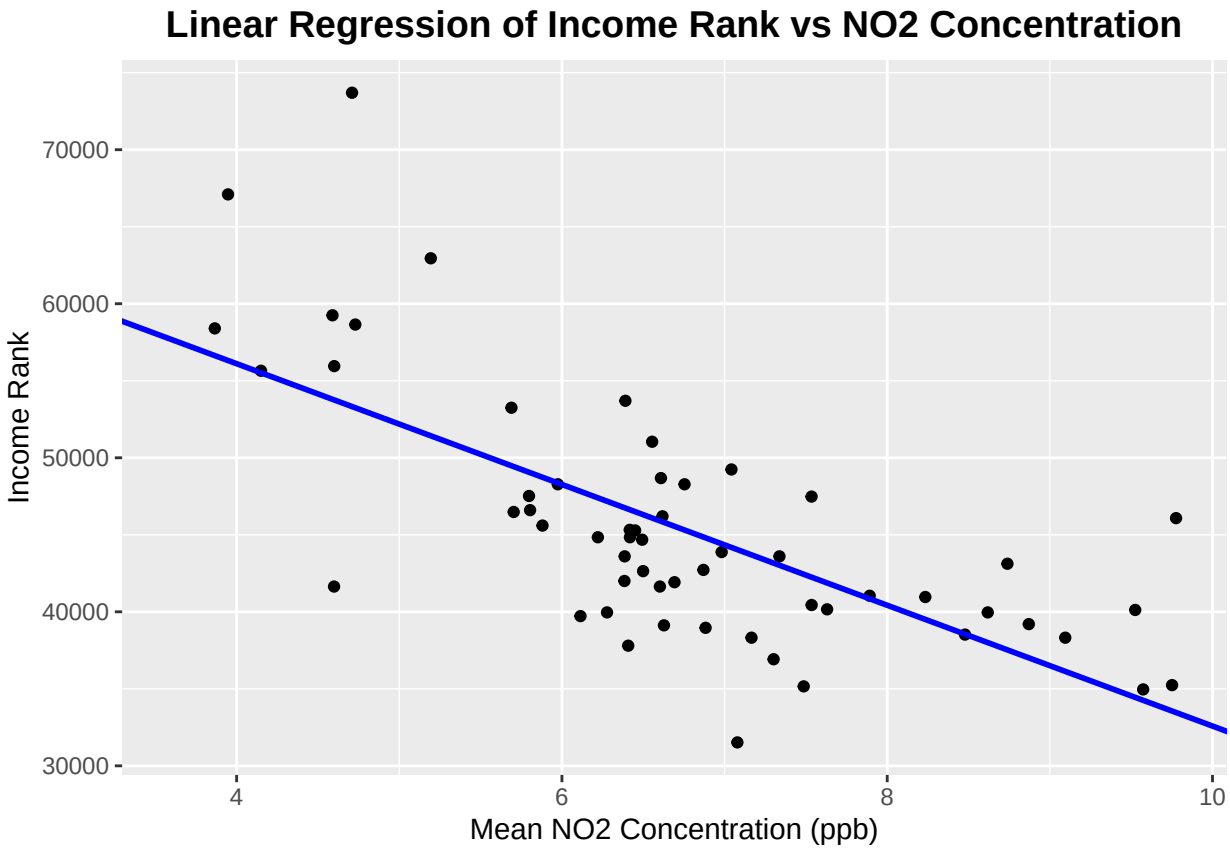



*Regression analysis*

Next, we will use a regression analysis to further investigate the relationships we have observed.

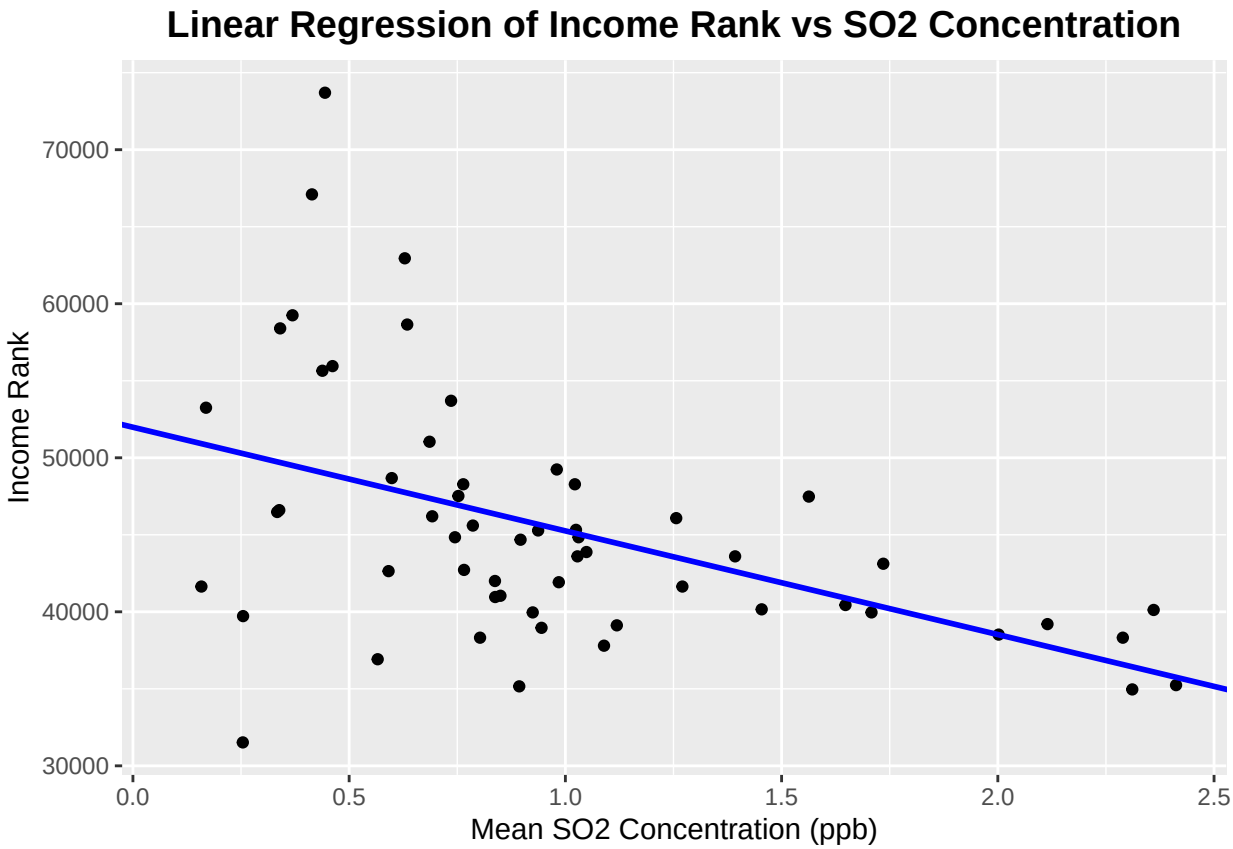
```
##
## Call:
## lm(formula = C1_COUNT_T ~ mean_NO2, data = HamDAs_WGScopy)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12515.0  -3152.5    -83.9   2801.7  20375.4
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  71789.3     3808.7  18.849  < 2e-16 ***
## mean_NO2     -3920.7      553.7  -7.081 2.78e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5970 on 55 degrees of freedom
##   (23 observations deleted due to missingness)
## Multiple R-squared:  0.4769, Adjusted R-squared:  0.4674
## F-statistic: 50.14 on 1 and 55 DF,  p-value: 2.777e-09

## Warning: Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



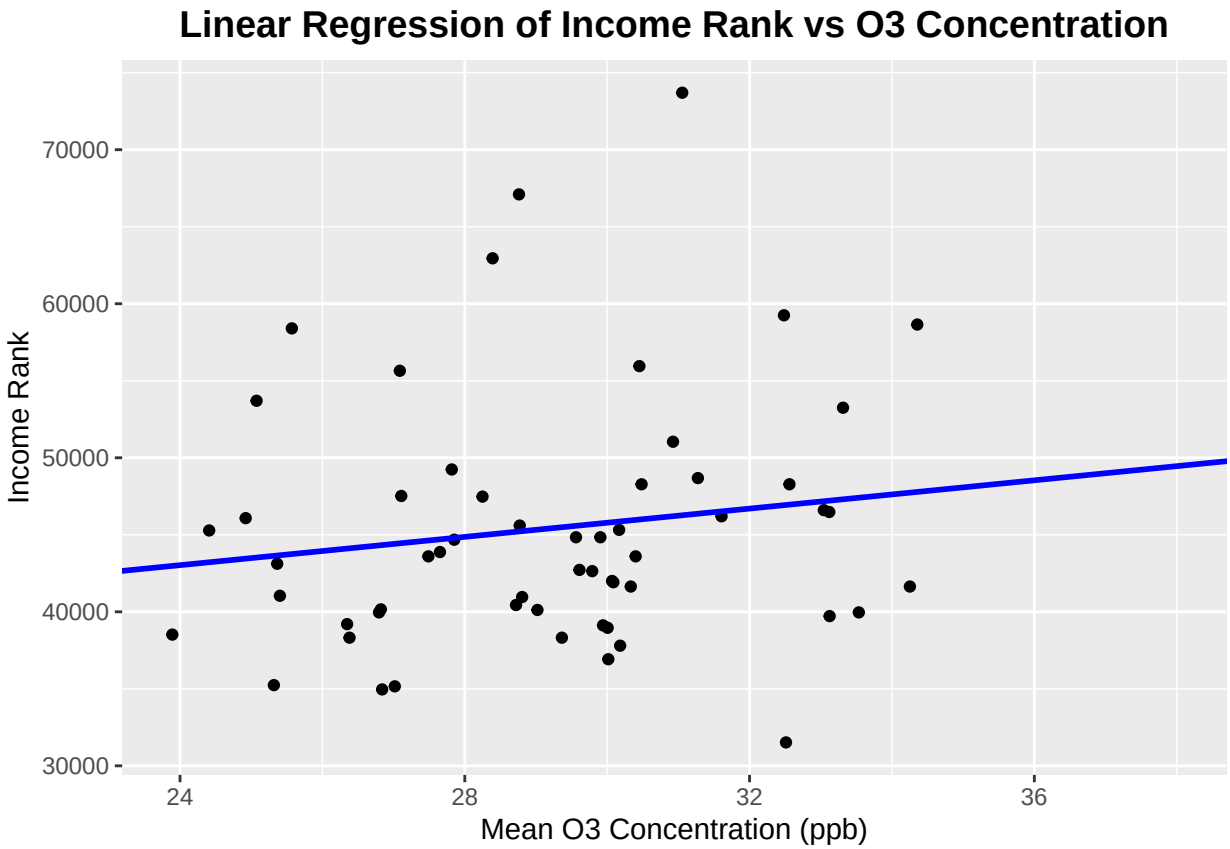

```
##
## Call:
## lm(formula = C1_COUNT_T ~ mean_SO2, data = HamDAs_WGScopy)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18751.9  -4109.1   -508.3   2815.6  24706.9
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    51982       1901  27.341  < 2e-16 ***
## mean_SO2       -6730       1679  -4.009 0.000185 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7262 on 55 degrees of freedom
##   (23 observations deleted due to missingness)
## Multiple R-squared:  0.2262, Adjusted R-squared:  0.2121
## F-statistic: 16.07 on 1 and 55 DF,  p-value: 0.0001851

## Warning: Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
##
## Call:
## lm(formula = C1_COUNT_T ~ mean_O3, data = HamDAs_WGScopy)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15415.7  -4905.1   -898.5   2501.7  27433.5
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  32007.7    12000.2   2.667    0.010 *
## mean_O3        459.2      409.5   1.121    0.267
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8162 on 55 degrees of freedom
##   (23 observations deleted due to missingness)
## Multiple R-squared:  0.02234,    Adjusted R-squared:  0.004568
## F-statistic: 1.257 on 1 and 55 DF,  p-value: 0.2671

## Warning: Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




We can map the residuals that arise from the linear regression in order to dissect the model we have produced's accuracy.

```
##
##  Moran I test under randomisation
##
## data:  HamDAs_WGS_valid$model_NO2
## weights: HamDAs_WGS.w
##
## Moran I statistic standard deviate = 4.1688, p-value = 1.531e-05
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##        0.31951368       -0.01785714        0.00654919
```

```
##
##  Moran I test under randomisation
##
## data:  HamDAs_WGS_valid$model_SO2
## weights: HamDAs_WGS.w
##
## Moran I statistic standard deviate = 6.5066, p-value = 3.844e-11
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##       0.505799631      -0.017857143       0.006477183
```

```
##
```

```
##  Moran I test under randomisation
##
## data:  HamDAs_WGS_valid$model_O3
## weights: HamDAs_WGS.w
##
## Moran I statistic standard deviate = 7.6538, p-value = 9.756e-15
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##       0.600097222      -0.017857143       0.006518652
```

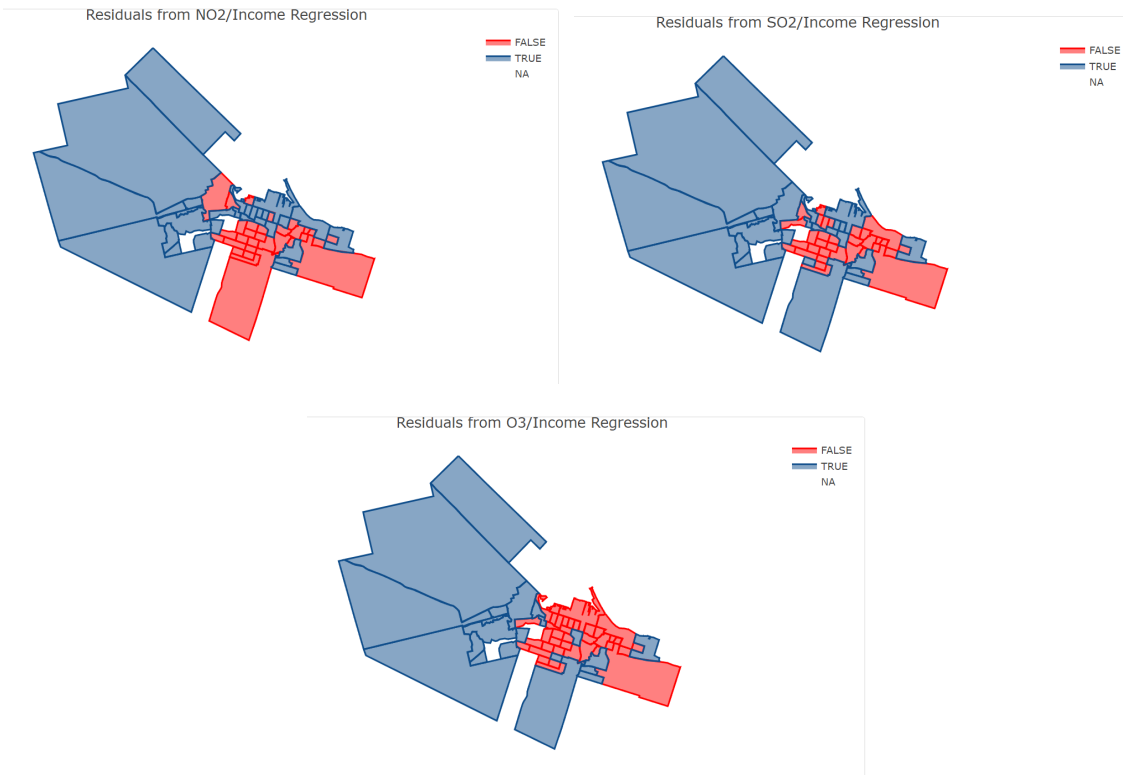



Figure 1: Image 3

## Results

*Spatial Autocorrelation (Moran's I)*

The Moran's I tests revealed that $NO_2$, $SO_2$, and $O_3$ all displayed a significantly significant spatial autocorrelation, with all three tests exhibiting p-values all less than 2.2e-16. $NO_2$ and $SO_2$ (seen in Moran's I Plots) had Moran's I values of 0.780 and 0.752, respectively. This indicates strong positive spatial clustering, meaning pollutant concentration values tend to be surrounded by similar pollutant concentration values. $O_3$ had a Moran's I value of 0.620, also indicating positive spatial autocorrelation, though slightly weaker than the other two pollutants. In contrast, the Moran's I test for income exhibited a test statistic of 0.039 and a p-value of 0.076, which is not statistically significant. This suggests that income distribution across dissemination areas is more spatially random compared to pollutant concentrations.

*Pearson's Correlation Analysis*

Pearson's correlation tests exhibited statistically significant values for both $NO_2$ and $SO_2$, with p-values of 2.78e-09 and 0.0001851, respectively (seen in Pearson correlation plots). This suggests that lower income levels are significantly correlated with higher pollutant concentration levels. The correlation coefficients were -0.69 for $NO_2$ and -0.48 for $SO_2$, confirming that as pollutant levels increase, income tends to decrease. The Pearson correlation for $O_3$ exhibited a p-value of 0.267, meaning that $O_3$ concentration does not have a meaningful correlation with income in Hamilton.

*Linear Regression Analysis*

Linear regression analyses were conducted to add to the rigour of our method in exploring how each pollutant predicted income levels. The linear regression results for $NO_2$ exhibited a p-value of 2.78e-09 and an $R^2$ of 0.477 (seen in Linear Regression plots), further suggesting a significant negative association between $NO_2$ concentration levels and income. Meaning as income levels decrease, $NO_2$ concentration increases. The findings for $SO_2$ were similar to that of $NO_2$, also showing a significant negative association, resulting in a p-value of 0.0001851 and an $R^2$ of 0.226. This result further confirms that as $SO_2$ concentration increases, income levels decrease. Linear regression results for $O_3$ displayed a p-value of 0.267 and an $R^2$ of 0.022, meaning that there is a not a statistically significant correlation between $O_3$ pollutant concentration levels and income levels. The findings from the linear regression analysis support those of the Pearson's correlation tests.

*Residuals and Model Fit*

Residuals analysis of the linear regressions were conducted using Moran's I to test the model fit. The residuals analysis for $NO_2$ exhibited a Moran's I test statistic of 0.320 and a p-value of 1.53e-05. The residuals analysis for $SO_2$ exhibited a Moran's I test statistic of 0.506 and p-value of 3.84e-11. Lastly, the residuals analysis for $O_3$ exhibited a Moran's I test statistic of 0.600 and a p-value of 9.76e-15. These results indicate spatial autocorrelation in the residuals, meaning they are not randomly distributed. This suggests that the regression models do not fully capture the spatial structure in the data. This indicates the presence of spatial processes not accounted for in the linear regression, suggesting that another method may need to be used to have a better model fit.

## Discussion

*Pollutant Concentrations*

This study analyzed air quality data from Open Hamilton and income status data from Statistics Canada. To compare the air quality data with income status, one mean concentration for each chemical was derived per dissemination area. An interpolated surface of each chemical using inverse distance weighting was made. When assessing the initial NO2 and SO2 concentrations in Hamilton, the highest concentrations appear to be clustered towards the central region of the city. There is a particularly high concentration in Downtown Hamilton, which is more proximate to Hamilton Harbour. Historically, these would have been the regions where the industrial sector would have flourished, known as the Bayfront Industrial Area (Merriman, 2011). Studies have also found there to be generally 50-100% higher concentrations of SO2, NO2, and CO at a site near a steel plant (Liu et al., 2014). Our analysis corroborates these initial observations and information from previous literature.

When assessing the asthmatics concentrations, the lower concentrations cluster towards the harbourfront region. This phenomenon may occur due to the chemical reactions that occur between O3 and NO2 (Ministry of the Environment, Conservation and Parks, 2010). As a major component of smog, O3 is not directly emitted as a contaminant, instead, it is a secondary pollutant produced from photochemical reactions between nitrogen oxides (NOx) and volatile organic compounds (VOC). Large point sources, such as a steel plant, generate large emissions of NOx and very little VOC, which prevents significant O3 formation. Furthermore, a process known as NOx titration occurs at nighttime in the immediate vicinity of very large emissions of

NO (Jhun et al., 2015). This occurs because freshly emitted NO reacts rapidly with O3 to produce NO2. At nighttime, there is no O3 formation and loss through NOx, therefore causing a lower O3 trend (Sillman, 1999). Studies have found decreased O3 within 80 km of a plume source (Sillman, 1999). O3 depression also tends to occur downwind of these large point sources, which supports the increasing O3 trend across Downtown Hamilton when moving southward.

*Comparisons with Null Landscapes and Spatial Autocorrelation Results*

When comparing the SMA of observed and simulated landscapes of NO2 concentrations across the City, the industrial centre has a clear concentration of higher NO2 in the empirical data. The simulated maps have scattered patterns without a clear hotspot. The conclusion can be made that the spatial distribution of NO2 is not random. This is further supported by the scatterplot of raw NO2 concentrations. There is a strong positive linear relationship, indicating that DAs with high NO2 are surrounded by other DAs with similar values. The same can be said for DAs with low NO2 values. Moran's statistics were also performed based on income statistics, which reveal a trend of environmental inequality. The results of the NO2 and SO2 Moran's plots confirm the previous conclusion, with an upward trend indicating that higher values of the two chemicals are surrounded by other high values, and vice versa. This suggests the concentrations of these pollutants are not randomly distributed, but spatially clustered. The O3 Moran's plot also has a positive relationship, however, the data points appear more scattered than the previous two chemicals. This suggests there may be more factors that impact the concentration of O3, such as potentially edge effects from the previously discussed downwind O3 transportation process.

Based on the results from the Moran's plots on both income and pollutants, we can assume that there is a process behind the pattern we observe in both income status and in the concentration of pollutants across the city. In order to find that process, we used correlation tests to see if these variables were related.

*Relationship between Pollutant Concentration and Income*

The lowest-income DAs tend to be concentrated near the industrial core of Hamilton. These coincide with the regions with the higher concentrations of NO2 and SO2 based on the results from our analysis. In contrast, the high-income DAs appear to be distanced from the industrial core, and therefore distanced from higher concentrations of pollutants, often situated in regions such as Ancaster, Dundas, or further south of Downtown Hamilton. A societal explanation for this trend could be due to the historical suburban redlining in Hamilton. In the 1930s, when Hamilton first began growing as an industrial city, factories expanded along the waterfront and residential land use developed parallel to the east (Harris & Forrester, 2003). However, the least desirable locations have always been closer to the harbourfront, where the industrial development dominated. In contrast, the most desirable locations were further south, moving towards the foot of the Niagara escarpment. A 1951 assessment record indicated that one-third to two-fifths of the male household heads who resided in the south and south-eastern regions of Hamilton were in professional or managerial positions (Harris & Forrester, 2003). This phenomenon could be due to the process of redlining, which refers to when institutions decline to make mortgage loans in specific areas. This process has historically been associated with disinvestment in the inner city. Studies revealed redlining in Hamilton sharply discriminated against the east end of the city and regions near the lakefront industry. Neighbourhoods in the west end, near the Niagara Escarpment and Bartonville, were favoured by insurance companies (Harris & Forrester, 2003).

Ultimately, the results of this study suggest there is a strong correlation between lower-income demographics and regions of the city with higher concentrations of pollutants, particularly NO2 and SO2. Furthermore, a history of discriminatory policies appears to have exacerbated the issue, preventing lower-income demographics from relocating to regions with more optimal environmental qualities.

*Limitations*

The Hamilton Air Monitoring Network (HAMN) is an industry-funded initiative composed of 22 local companies that contribute to the ongoing monitoring of ambient air quality in the City of Hamilton (City of

Hamilton, 2024). While this network provides valuable data, several limitations must be acknowledged. First, the spatial distribution of air quality monitors accessed through open data sources is uneven across the study area. In some regions, monitors are clustered within a small radius, whereas other areas are represented by a single monitor covering a considerably larger geographic extent. This uneven distribution may introduce spatial bias and result in data gaps, thereby reducing the accuracy and representativeness of air quality assessments.

Second, there is limited publicly available information regarding the calibration frequency and maintenance schedules of these monitoring stations. Inadequate or infrequent calibration can compromise data reliability, potentially leading to either inflated or underestimated pollution measurements.

Third, the siting of monitoring stations may influence the recorded pollutant concentrations. Monitors are situated in varying environments, ranging from areas adjacent to major transportation corridors and industrial facilities to locations surrounded by vegetation or open space. Such variability in site context can significantly impact pollutant readings. In addition, meteorological events such as wind direction play a critical role in pollutant dispersion and monitoring readings (Coker, 2024). For example, in industrial zones, prevailing winds may direct emissions toward a nearby monitor, producing results that are not representative of broader regional conditions.

Finally, a notable limitation involves the unavailability of health data due to privacy and confidentiality constraints. This restricted access to health-related datasets hindered the ability to explore correlations between air quality and public health outcomes, such as asthma incidence, cardiovascular conditions, or hospital admissions for respiratory illnesses.

*Current and Future Initiatives*

The governments of Canada and Ontario have regulations in place to help mitigate the effects of emissions in Hamilton. Air quality plays a critical role in human health and well-being, directly impacting quality of life and day-to-day activities (City of Hamilton, 2024). In Hamilton, steel plants are required to measure and report their emissions annually to the Ontario Ministry of the Environment, Conservation and Parks (MECP) as part of Ontario's commitment to holding large industrial emitters accountable for their contribution to total greenhouse gas emissions. This reporting is a key component of Ontario's Emissions Performance Standards (EPS) program, which provides verified data on emissions, production, and emission limits for all producers. The data is used to determine compliance, issue air quality warnings, and allocate Emission Performance Units (EPUs) to companies based on their reduced emissions (Government of Ontario, 2014). EPUs are tradable units given to facilities that keep their emissions below set limits. These units act as an economic policy tool, creating financial incentives for industries to reduce their GHG emissions and fostering a market for emission reduction.

The Ontario government also has Regulation O. Reg. 419/05: Air Pollution - Local Air Quality, under the Environmental Protection Act of 1990. On a local level, this regulation protects communities from the effects of air pollution by regulating air contaminants released from nearby sources. Provincial air standards set by Ontario's Local Air Quality Regulation are science-based limits designed to protect human health and the environment (Burdon, 2023). These standards apply to contaminant concentrations outside a facility's property boundary and are used to assess emissions and identify future management areas (Burdon, 2023). When these standards cannot be met, alternative compliance options, such as site-specific standards, may be considered (Burdon, 2023).

As part of this regulation, every new site-specific standard request requires a fully refined Emission Summary and Dispersion Modelling (ESDM) Report (Burdon, 2023). This document demonstrates the facility's ability to comply with air limits or site-specific standards and includes details on facility operations, the types and quantities of air emissions, and the results of computer modelling used to estimate pollutant concentrations beyond the facility's property boundary (Burdon, 2023). The report helps government officials assess whether the facility can meet air quality standards, identify potential risks to public health and the

environment, and guide the development of strategies to minimize harmful emissions.

The governments of Canada and Ontario are partnering with ArcelorMittal Dofasco to support the company's transition to cleaner steel production technologies. Together, they have committed $900 million to the implementation of Direct Reduced Iron-Electric Arc Furnace (DRI-EAF) technology in Hamilton (ArcelorMittal Dofasco, 2022). This collaborative project is a major step in advancing decarbonization efforts this decade and reflects a shared commitment to reducing greenhouse gas emissions in the industrial sector. The DRI-EAF process replaces traditional coal-powered blast furnaces with a two-step system: direct reduction of iron to produce a low-carbon feedstock, followed by the use of an electric arc furnace (EAF) to melt the reduced iron using electricity (ArcelorMittal Dofasco, 2022). This significantly lowers emissions compared to conventional methods. The project supports ArcelorMittal's broader goal of reducing the carbon intensity of its steel production by 25% by 2030 (ArcelorMittal Dofasco, 2022).

The governments of Canada and Ontario can work together with various industrial and commercial sectors in Hamilton to develop funding programs that promote the adoption of cleaner, more sustainable alternatives across the city. This could include stricter regulations on emission limits, as well as incentives for businesses to implement adaptive measures and invest in sustainable technologies that reduce GHG emissions. These efforts might focus on enhancing energy efficiency, transitioning to renewable energy sources, and providing financial research support. Collaboration between government and industry is essential not only to reduce emissions but also to create a green economy that supports long-term environmental sustainability, health and overall well-being in Hamilton.

## Conclusion

The City of Hamilton, commonly known as the "Steel City," experienced significant industrial growth following the Industrial Revolution, with steel production becoming the dominant industry. This rise in manufacturing attracted a growing population seeking employment opportunities. However, it also led to increased housing demand, rising living costs, and deepening poverty, particularly in the downtown core. As a result, lower-income populations became concentrated near industrial emission sources, raising health concerns due to elevated levels of harmful pollutants such as ($NO_2$) and ($SO_2$) are linked to respiratory, neurological, and cardiovascular issues. These combined social and environmental challenges underscore the need for more equitable and sustainable urban development.

This study aimed to determine the spatial relationship between low-income status and exposure to air pollutants like $NO_2$, $SO_2$, and $O_3$, emitted from steel mills in Hamilton, Ontario. Statistical analysis of both the Pearson correlation tests and linear regression analysis revealed significant negative correlations between income and both $NO_2$ and $SO_2$ levels, indicating that lower-income areas are exposed to higher concentrations of these pollutants. In contrast, $O_3$ showed no meaningful correlation with income. In addition, the $NO_2$ and $SO_2$ Moran's plots confirmed that the pollutants are spatially clustered and are not randomly distributed. The $O_3$ Moran's plot also confirmed this positive relationship, however, the data points are more scattered than the previous chemicals. Overall, the majority of the pollutants examined from these steel plants are negatively impacting the poor communities with lower income levels in the City of Hamilton.

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